

CHARACTERIZATION OF CRYSTAL EFFICIENCY:

Methodology, Corrections, and Results

JULY 2026

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1. DATA ANALYZED

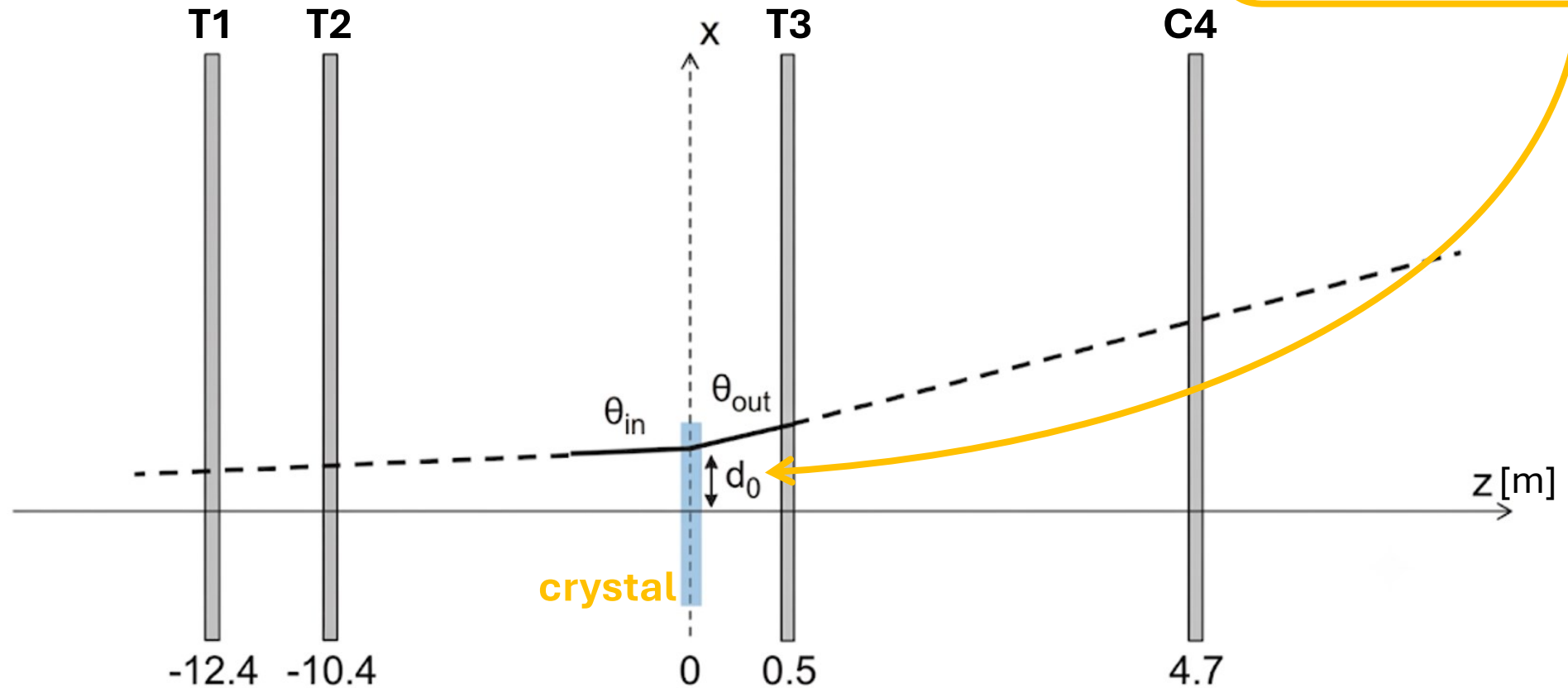
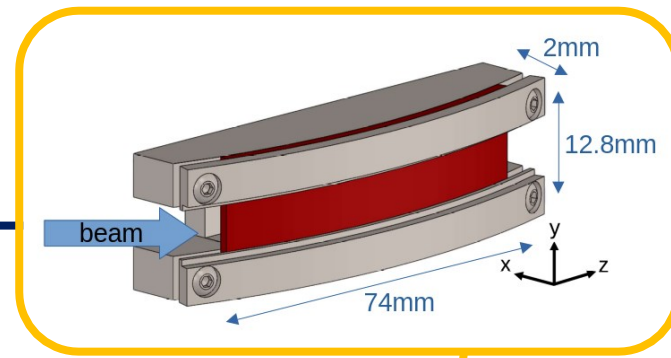
Analysis of 2024 dataset for two long crystals:

- LHCTBMCP0001-0001: runs 8430, 8431, 8650
- LHCTBMCP0001-0002: runs 8655, 8656

Configuration:

- Central goniometer
- Energy 180 GeV/c or 150 GeV/c for run 8650

1. SCHEMATIC LAYOUT



2. PREVIOUS ANALYSIS



SELECT SINGLE TRACKS

Selection of single tracks to eliminate multi-scattering events and multi-hit events (pile-up effect) from the dataset.



PARTICLE ENTRY

Validation of which particles successfully entered the crystal.

Since we do not know the relative beam position: algorithmic identification of the physical crystal edges based on interaction density profiles.



DETECTOR ALIGNMENT

Ensuring precision and the alignment of the detectors.

Any misalignment between tracking planes corrupts the angular measurements, necessitating corrections.

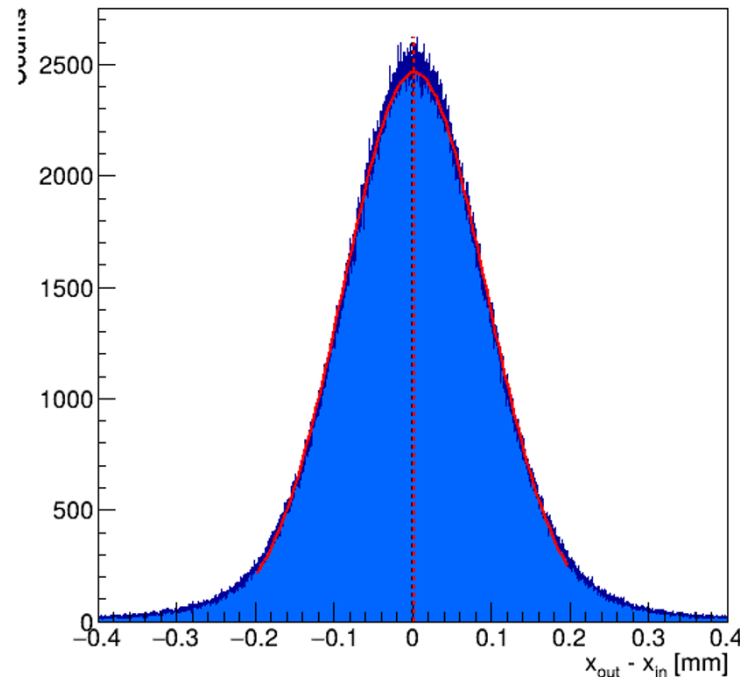
2. PREVIOUS ANALYSIS: ALIGNMENT

To ensure that the upstream and downstream detectors are aligned, we check the deviation between the initial and final positions of particles that definitely did not hit the crystal, ensuring that is zero.

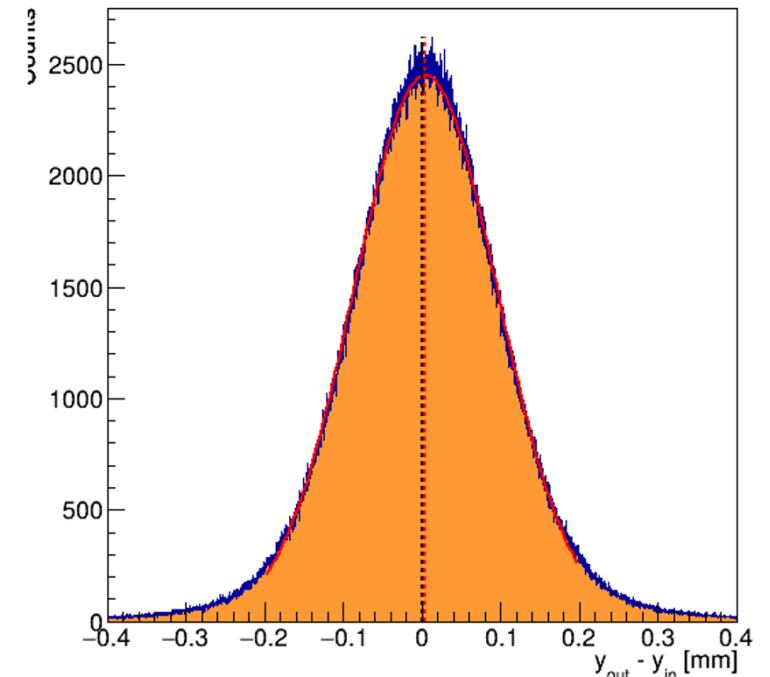
$\text{offset}_x = 0.003 \text{ mm}$
skewness ~ 0

$\text{offset}_y = 0.004 \text{ mm}$
skewness ~ 0

Alignment Offset X (Air Particles)

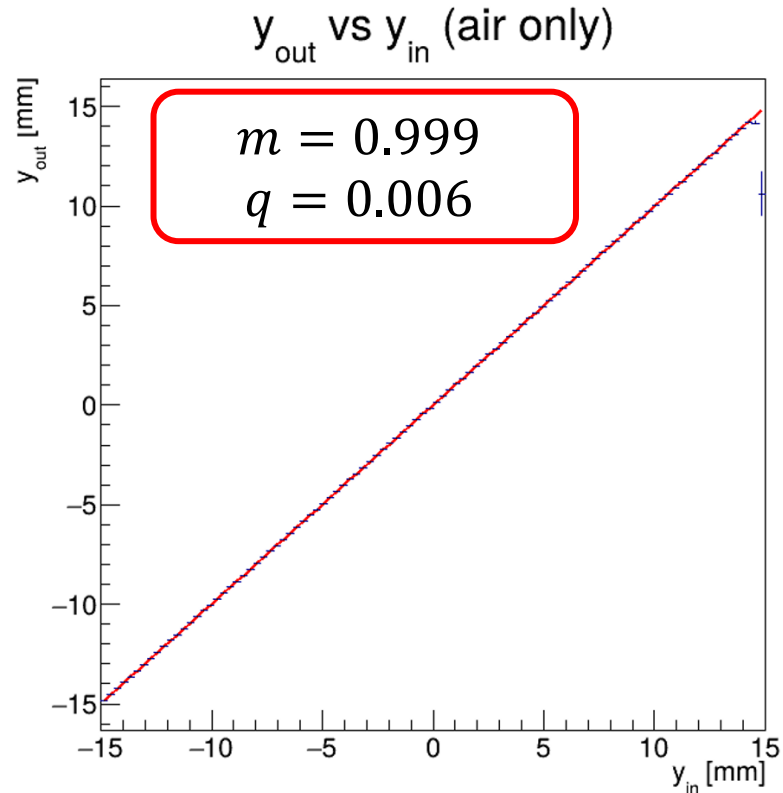
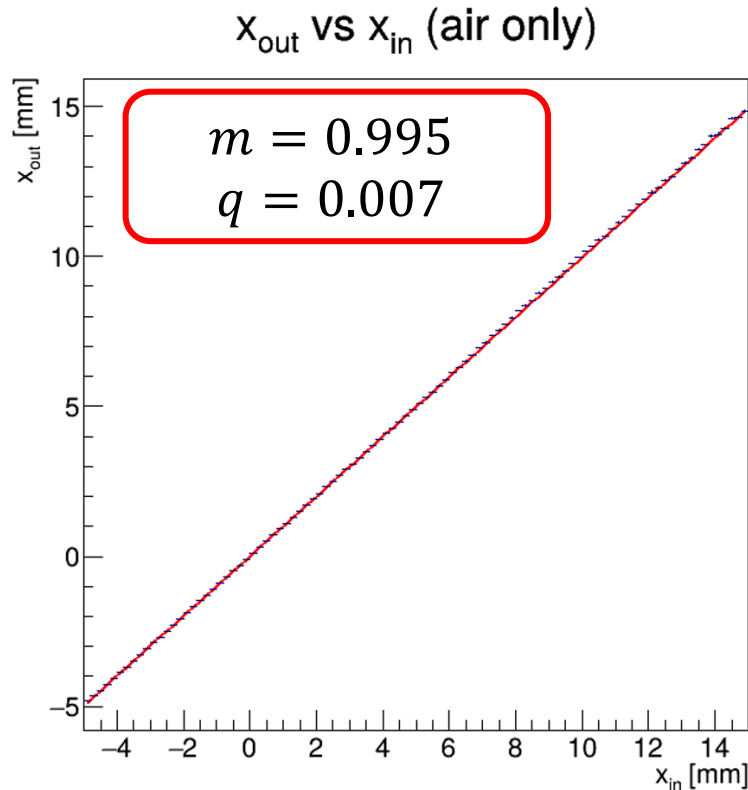


Alignment Offset Y (Air Particles)



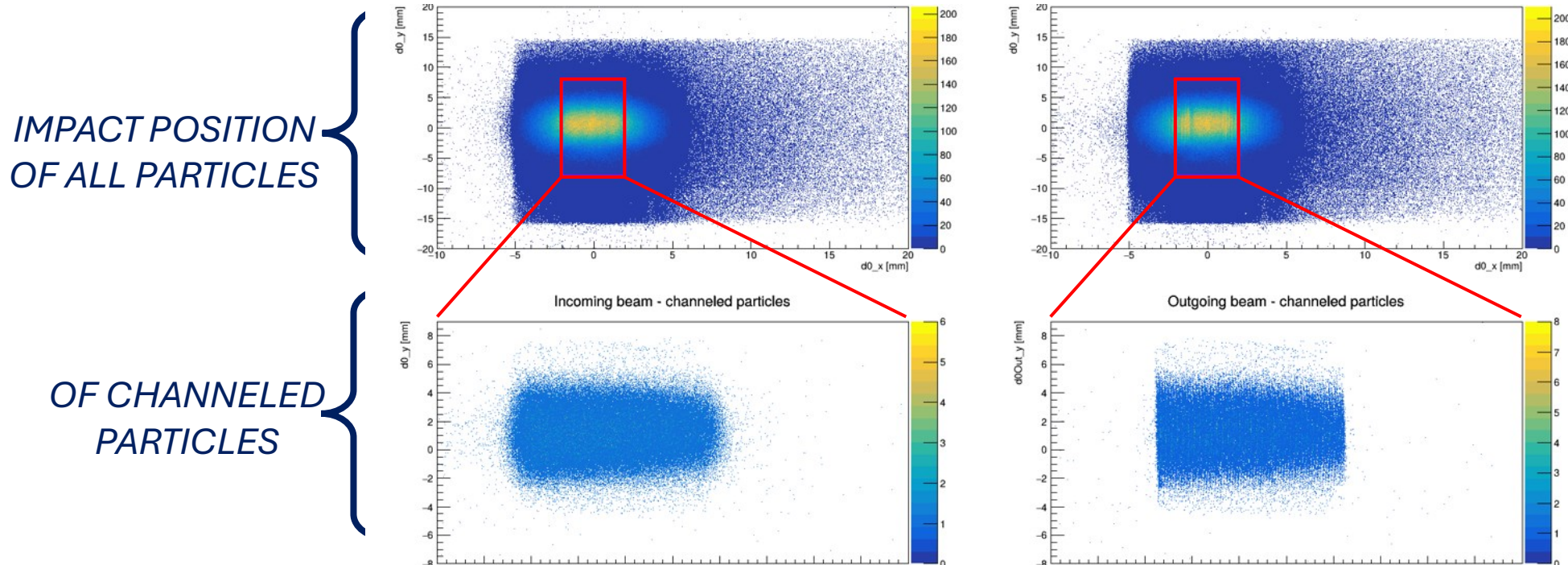
BACKUP SLIDES: ALIGNMENT

We can also check the alignment by doing a linear fit between the upstream and downstream profiles.



2. PREVIOUS ANALYSIS: CRYSTAL EDGES

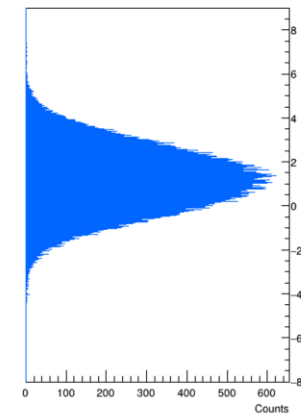
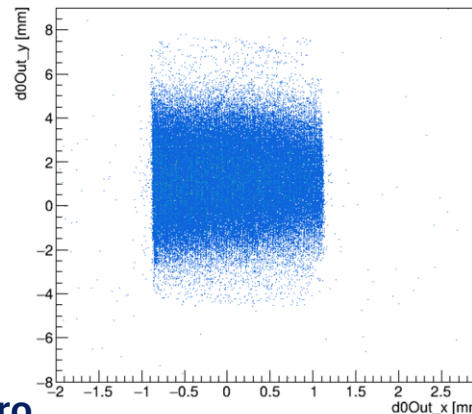
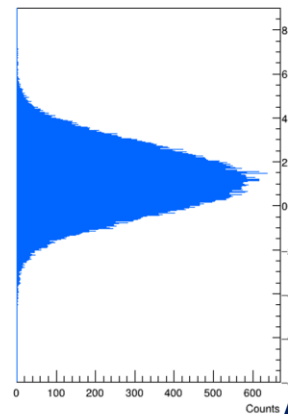
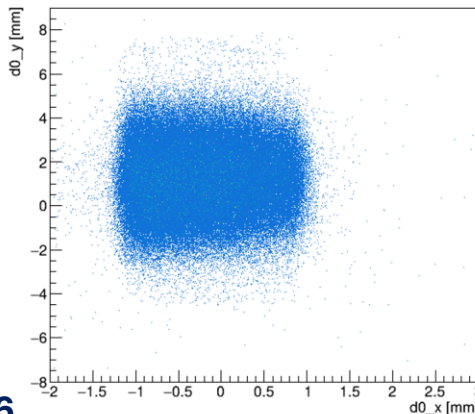
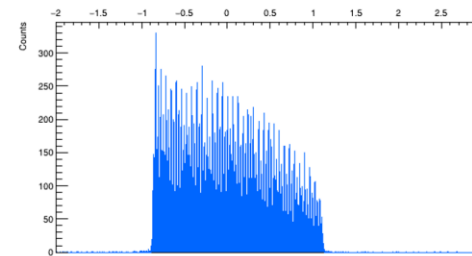
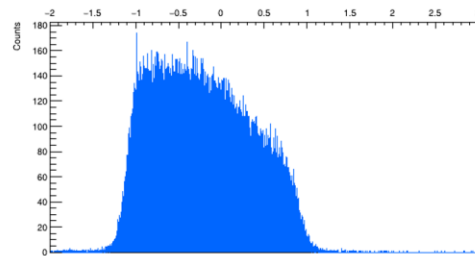
To select the particles that actually enter the crystal we first need to know the position of the crystal with respect to the beam.



We then look at channeled particles only using a preliminary deflection cut. Their impact points reveal the crystal's physical location, matching its expected dimensions.

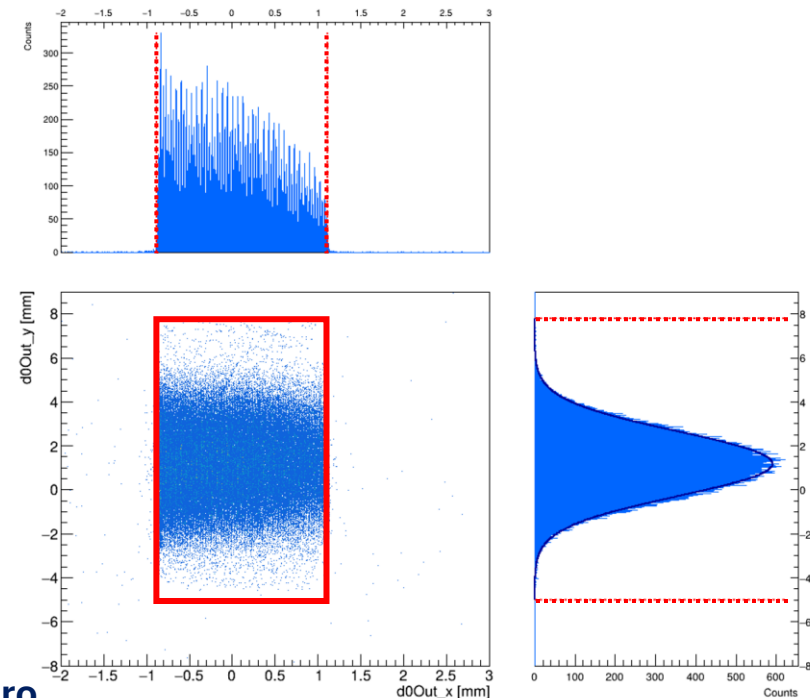
2. PREVIOUS ANALYSIS: CRYSTAL EDGES

- The impact points of these particles revealed the physical location of the crystal, matching its expected dimensions of 2 by 12.8 mm.
- To find the edges along x (that have an asymmetrical distribution) we perform a «sliding window» method (fixed width & precision of 0.01mm), aiming for the position that maximize the number of particles inside.



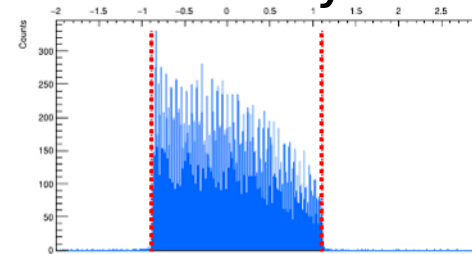
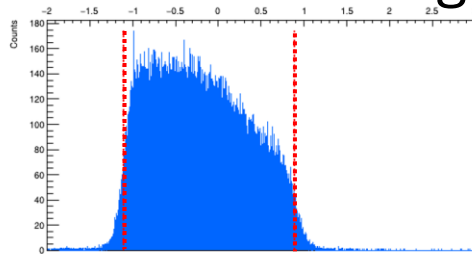
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- To find the edges along x (that have an asymmetrical distribution) we perform a «sliding window» method (fixed width & precision of 0.01mm), aiming for the position that maximize the number of particles inside.
- The method was performed on the downstream particles, since they show more defined edges.



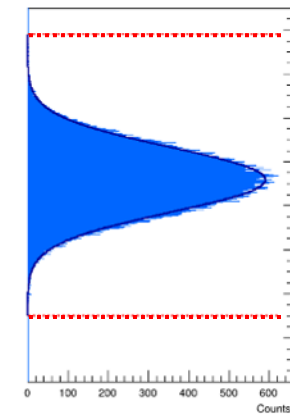
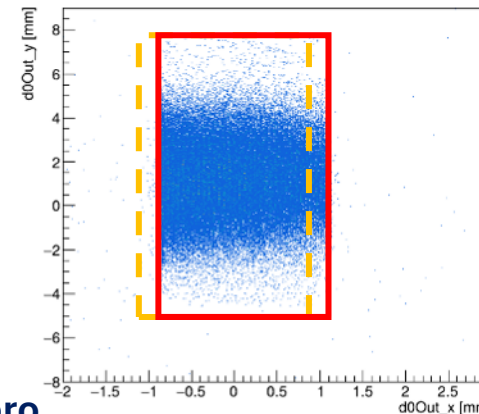
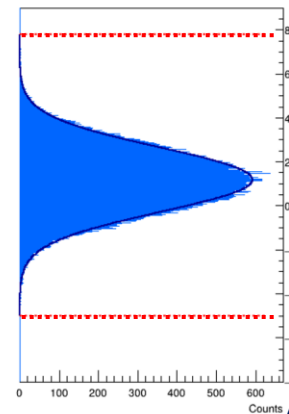
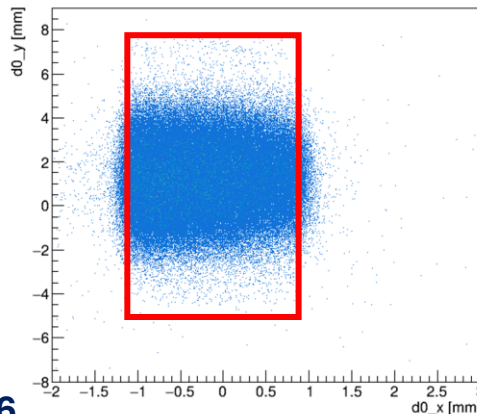
2. PREVIOUS ANALYSIS: CRYSTAL EDGES

- To define the true footprint at the entrance face we must project the downstream edges back upstream.
- We need to subtract a shift Δx from the downstream edge position.
- Channeled particles follow the bent lattice planes, resulting in a lateral geometric drift along the x -axis as they traverse the crystal.

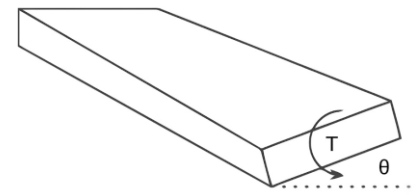


$$\Delta x = \frac{\theta_b \cdot L}{2}$$

$\sim 0.22 \text{ mm}$



3. ANALYSIS: TORSION EFFECT



Torsion effect: in long crystals (>1 cm), structural torsion is non-negligible and must be incorporated into the analysis.

- The different orientations of the crystallographic planes along the crystal front surface influence the angular range in which particles can be channelled.
- Before selecting the particles that have entered with the right angle (within $\pm \frac{1}{2} \theta_L$) we need to apply a correction θ_0 because the cut may not be centered at zero.
- We do not choose a single θ_0 value but rather $\theta_0(x, y)$ in order to also apply a torsion correction which is not negligible in the long crystals.

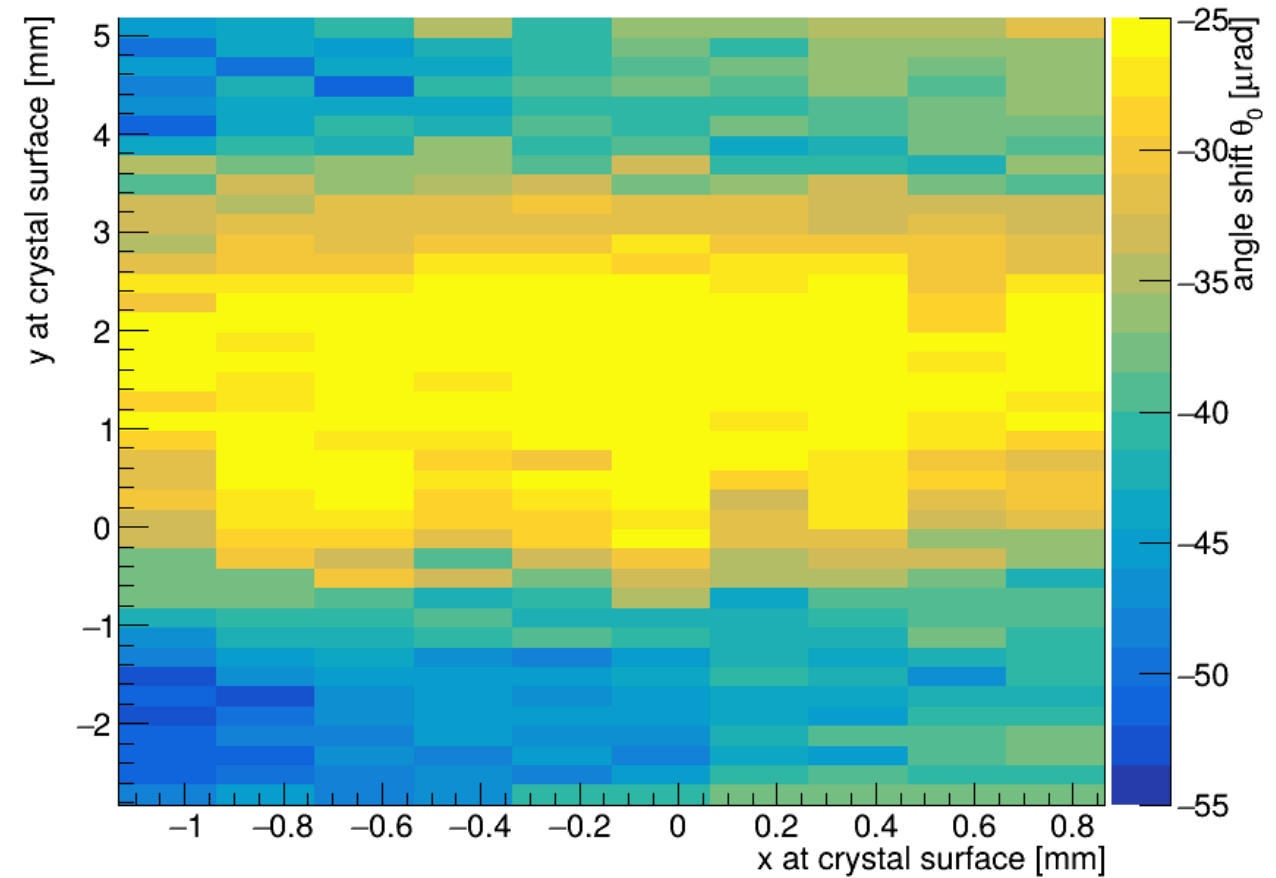
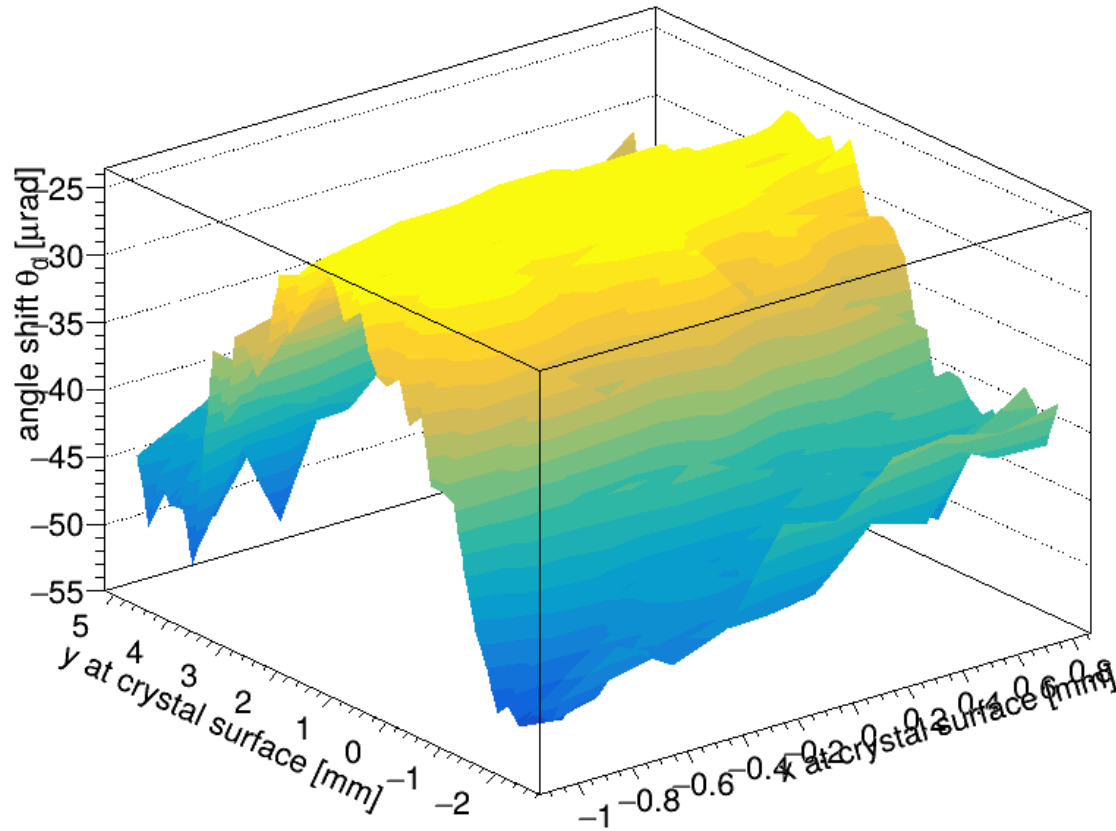
3. ANALYSIS: TORSION EFFECT

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The effect of the torsion can be corrected by doing a checkerboard analysis:

1. Dividing the data sample into square bins (for example 0.1 mm x 0.1 mm) as a function of the impact position on the crystal surface
2. The angular shift parameter θ_0 is then varied to find the maximum channeling efficiency ϵ_{ch} for each bin
3. We obtain a 2D map of torsion variation across the crystal surface

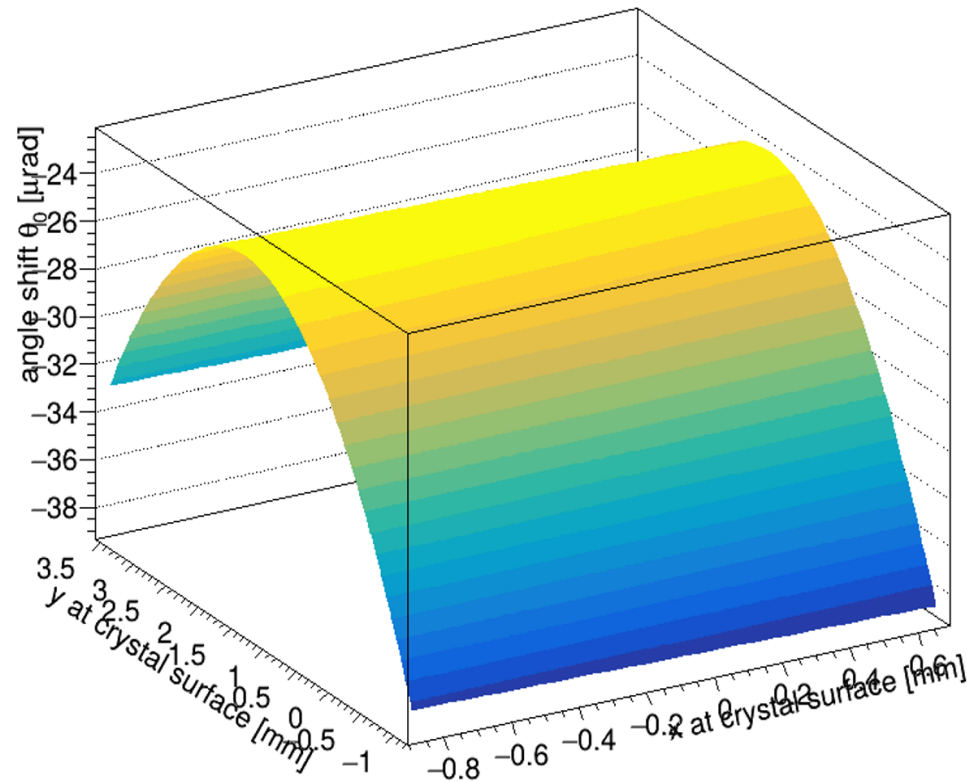
3. ANALYSIS: TORSION EFFECT



3. ANALYSIS: TORSION FIT

The 2d map is fitted through a parabolic function:

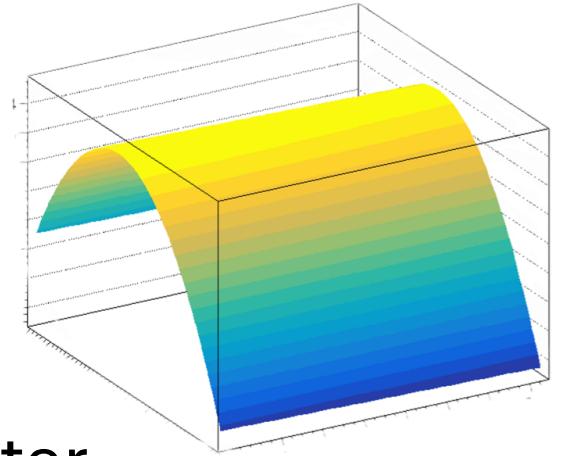
$$\theta_{corr}(x, y) = \theta_0 + \tau_x \cdot x + \tau_y \cdot y + p_3 y^2$$



3. ANALYSIS: TORSION EFFECT

The 2d map is fitted through a parabolic function:

$$\theta_{corr}(x, y) = \theta_0 + \tau_x \cdot x + \tau_y \cdot y + p_3 y^2$$



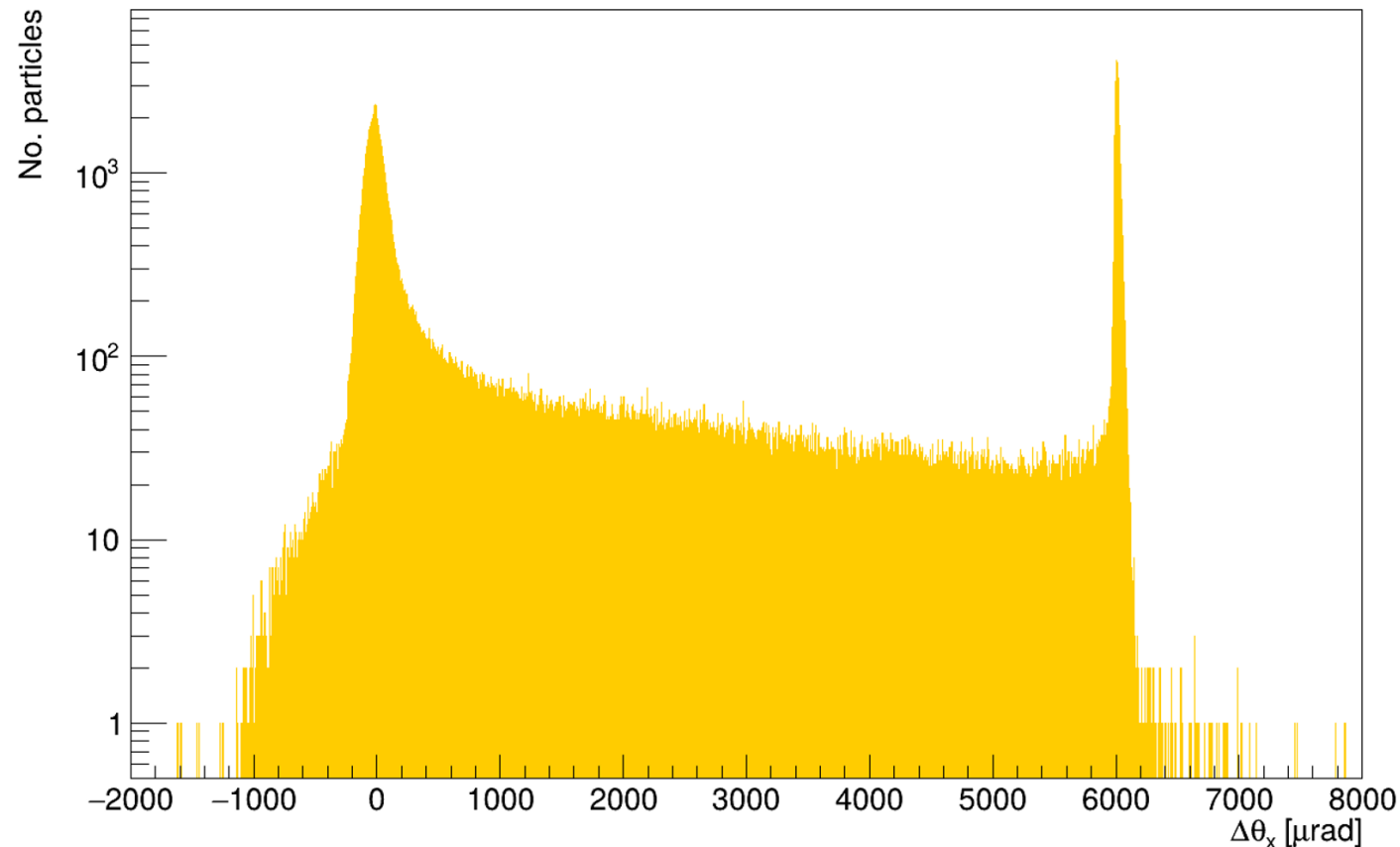
Now we correct the effect of torsion using the parameter $\theta_0(x, y)$ found for each bin.

instead of the Lindhard cut $|\theta_{in,x}| \leq \frac{1}{2} \theta_L$ we will use

$$\left| \theta_x^{in} - \left(\theta_x^{off} + \tau_y \cdot d_y + \tau_x \cdot d_x \right) \right| \leq \frac{1}{2} \theta_L$$

3. ANALYSIS

We can now compute the efficiency as the ratio of successfully channeled particles to the total number of incident particles.

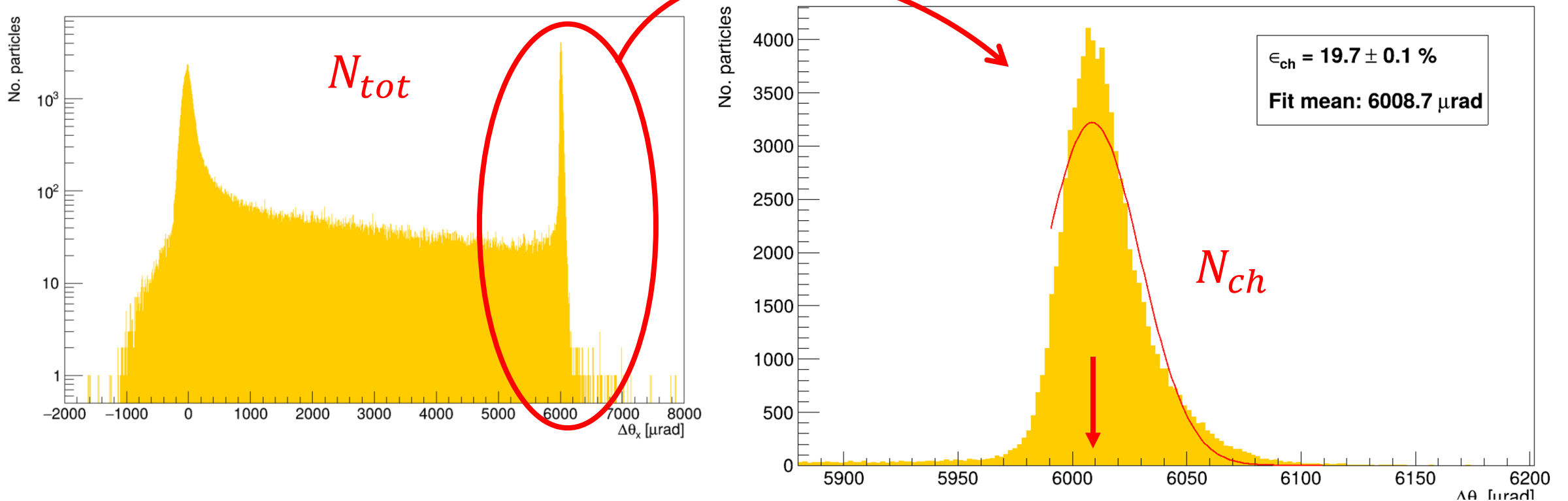


**CHANNELING
EFFICIENCY**

$$\epsilon_{ch} = \frac{N_{ch}}{N_{tot}}$$

3. ANALYSIS

- A Gaussian fit is performed on the deflection peak.
- We fit the right side of the peak to ensure the cleanest measurement (no dechanneled particles contribution).



4. RESULTS

	run	τ_y [$\mu\text{rad}/\text{mm}$]	θ_b [μrad]	ϵ_{ch} [%]
Long crystal 1	8430	6.25 ± 0.04	6008.4 ± 0.2	19.6 ± 0.01
	8431	5.71 ± 0.04	6008.4 ± 0.2	19.1 ± 0.01
	8650	3.89 ± 0.01	6081.7 ± 0.1	17.1 ± 0.01
Long crystal 2	8655	3.67 ± 0.02	6122.6 ± 0.3	18.0 ± 0.01
	8656	3.80 ± 0.01	6122.0 ± 0.3	18.0 ± 0.01

4. COMBINED RESULTS

	run	τ_y [$\mu\text{rad}/\text{mm}$]	θ_b [μrad]	ϵ_{ch} [%]
Long crystal 1	8430	5.98 ± 0.03	6008.4 ± 0.1	19.4 ± 0.01
	8431			
Long crystal 2	8655	3.77 ± 0.01	6122.3 ± 0.2	18.0 ± 0.01
	8656			

4. FINAL RESULTS

- Crystal 1 does show inconsistencies in different runs (8430 and 8650 have different energies)
- Crystal 1 shows a maximum efficiency of $\sim 25.4\%$
- Crystal 2 shows a maximum efficiency of $\sim 23.1\%$

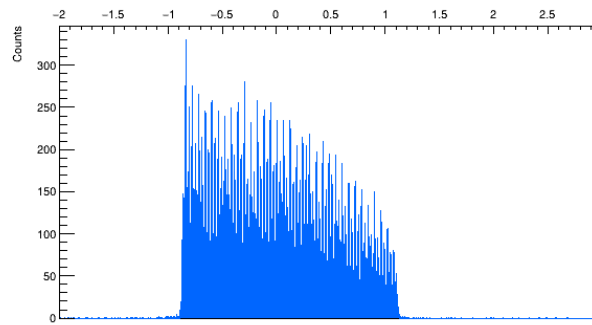
4. NEXT STEPS

ERRORS

Quantify statistical and systematic uncertainties on the channeling efficiency (ϵ_{ch}) and torsion parameters.

SPIKES

Explore the spikes found in the histogram from the channeled particles

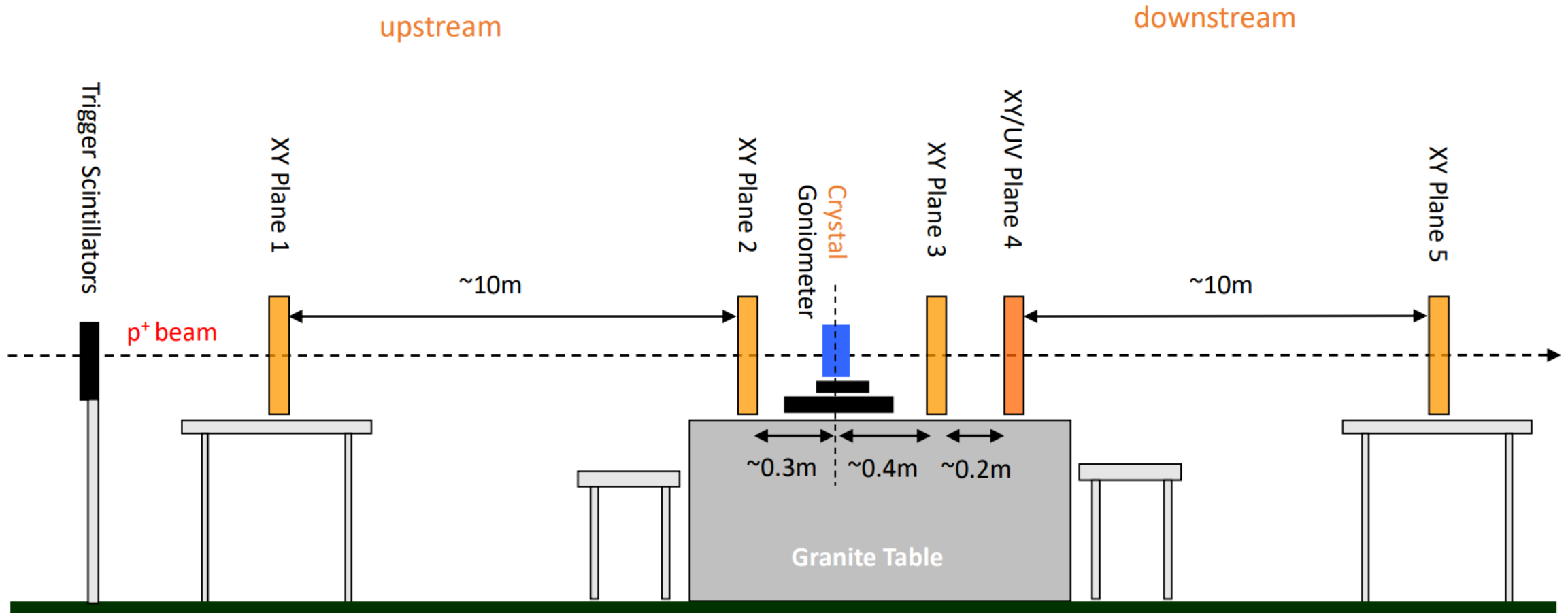


RUN 8650

Understand why run 8650 shows an incompatible channeling peak.

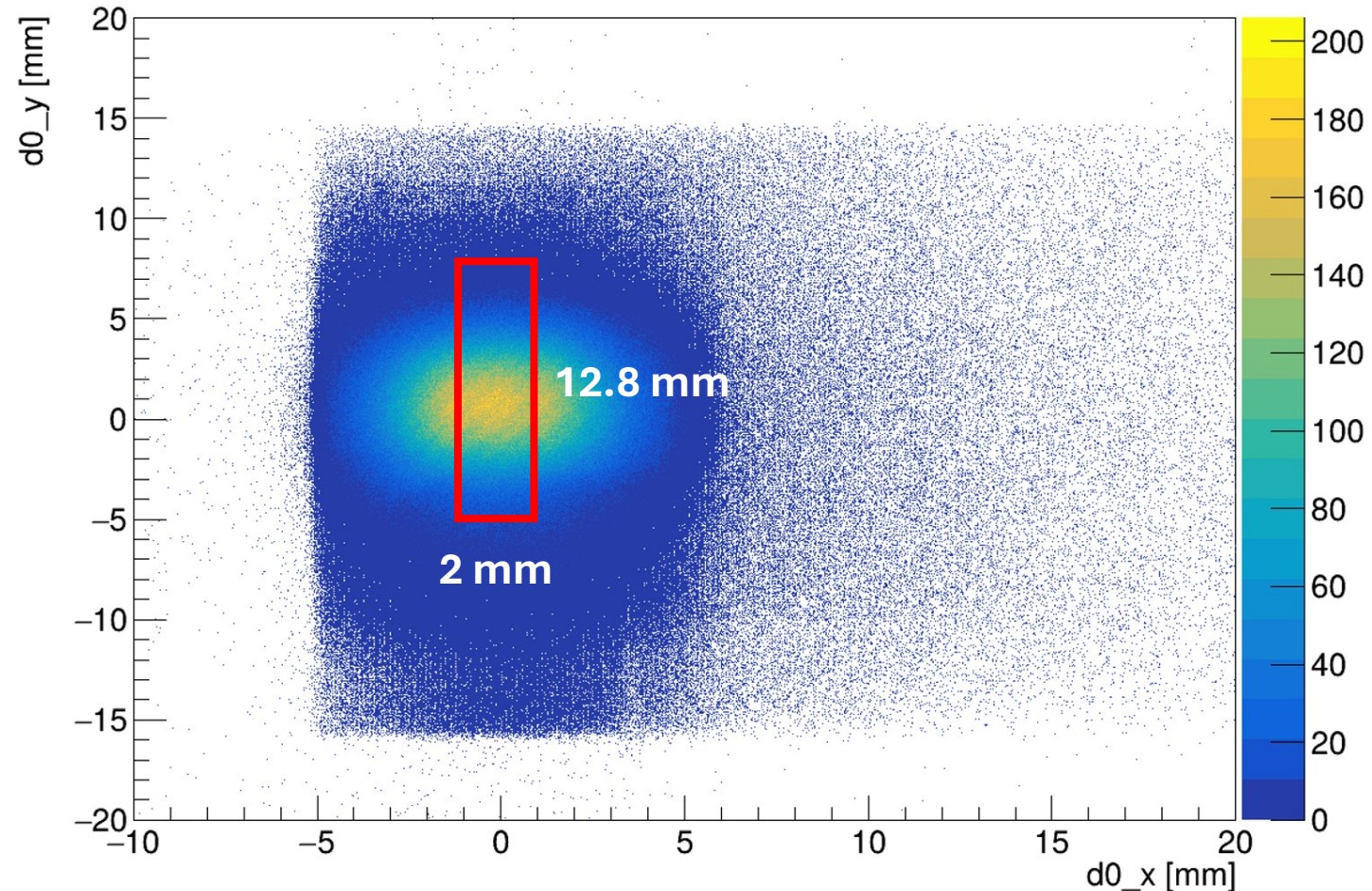
BACKUP SLIDES

BACKUP SLIDES: SCHEMATIC LAYOUT



BACKUP SLIDES: SPATIAL CUT

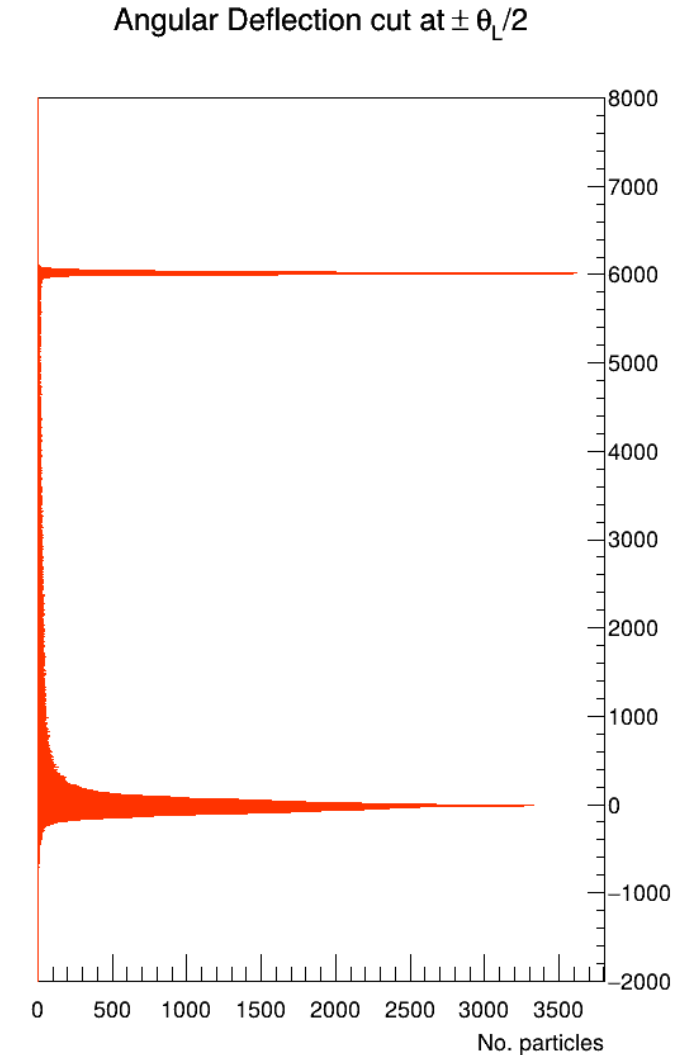
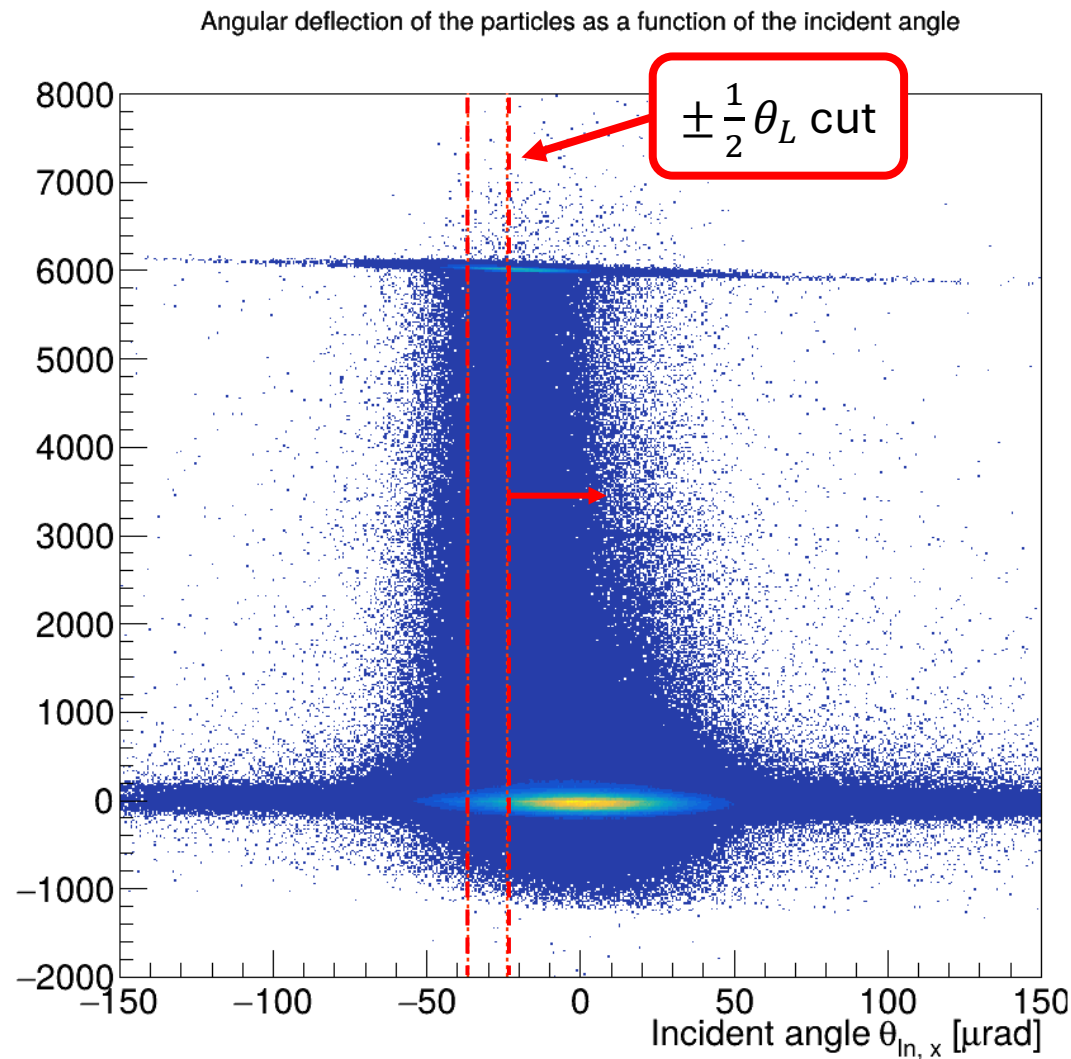
Incoming beam of all particles



BACKUP SLIDES: ANGLE SELECTION

For each square bin:

- variation of θ_0
- select particles
 $|\theta_{in} - \theta_0| \leq \frac{1}{2} \theta_L$
- compute ϵ_{ch}
- select θ_0 that
maximizes ϵ_{ch}
(through gaussian
fit)

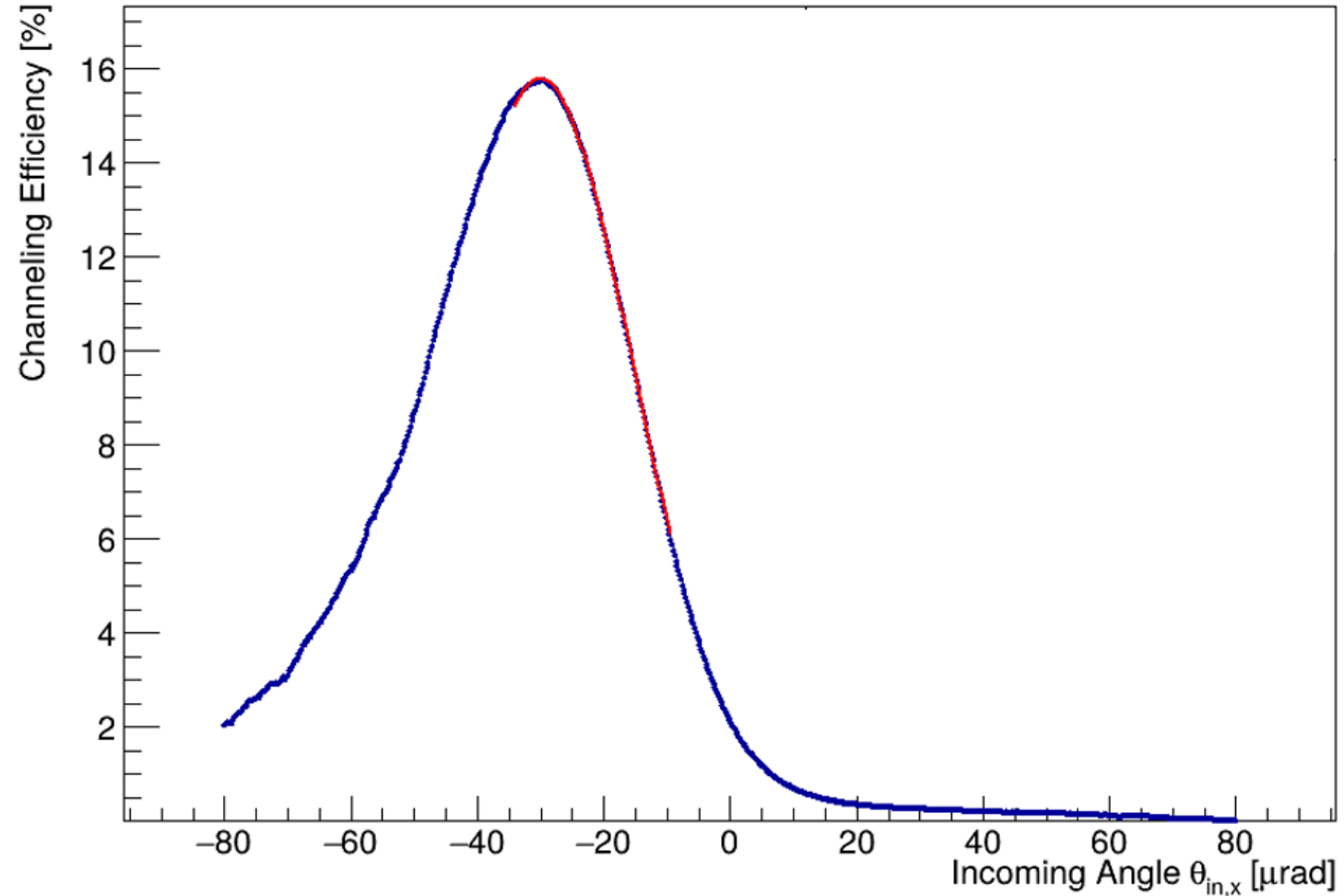


BACKUP SLIDES: ANGLE SELECTION

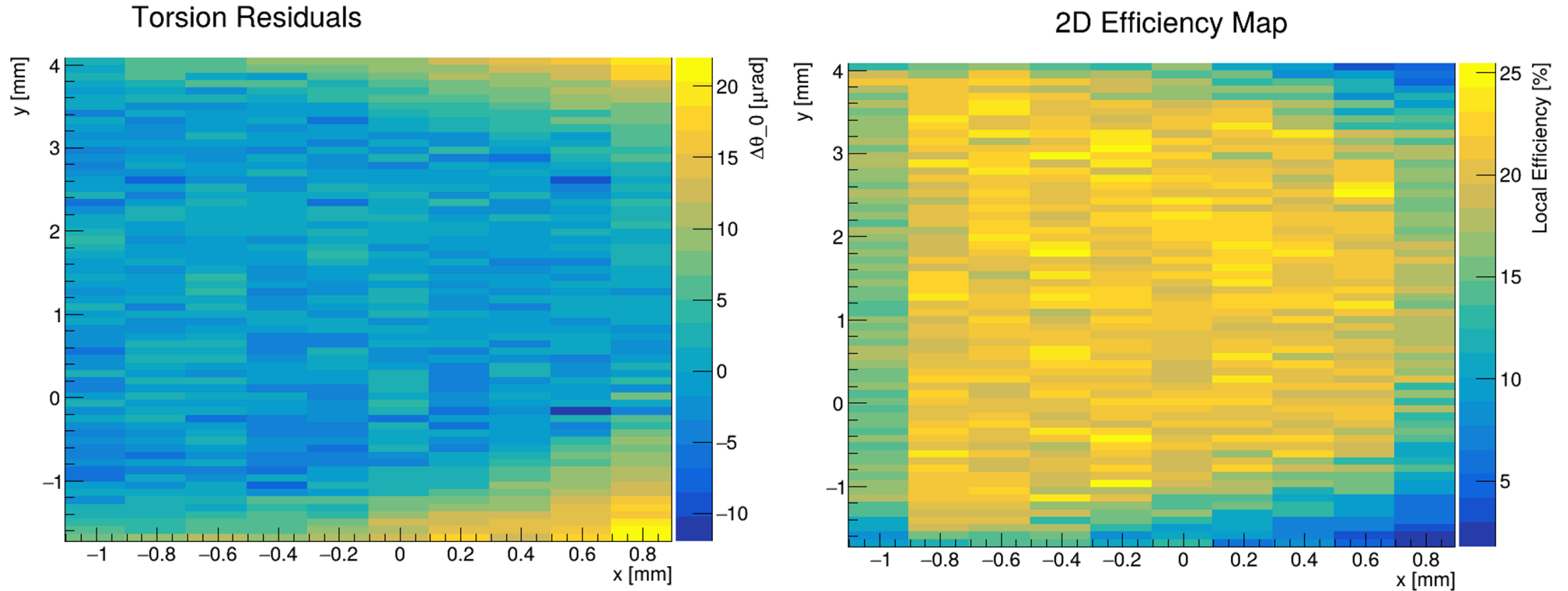
Angular Acceptance Curve

For each square bin:

- variation of θ_0
- select particles
 $|\theta_{in} - \theta_0| \leq \frac{1}{2} \theta_L$
- compute ϵ_{ch}
- select θ_0 that
maximizes ϵ_{ch}
(through gaussian
fit)



BACKUP SLIDES: TORSION FIT



BACKUP SLIDES: EFFICIENCY CURVE

Crystal efficiency as a function of the incoming angle, after application of the torsion correction. The value of the peak is the efficiency of the crystal when it is perfectly aligned to the beam.

